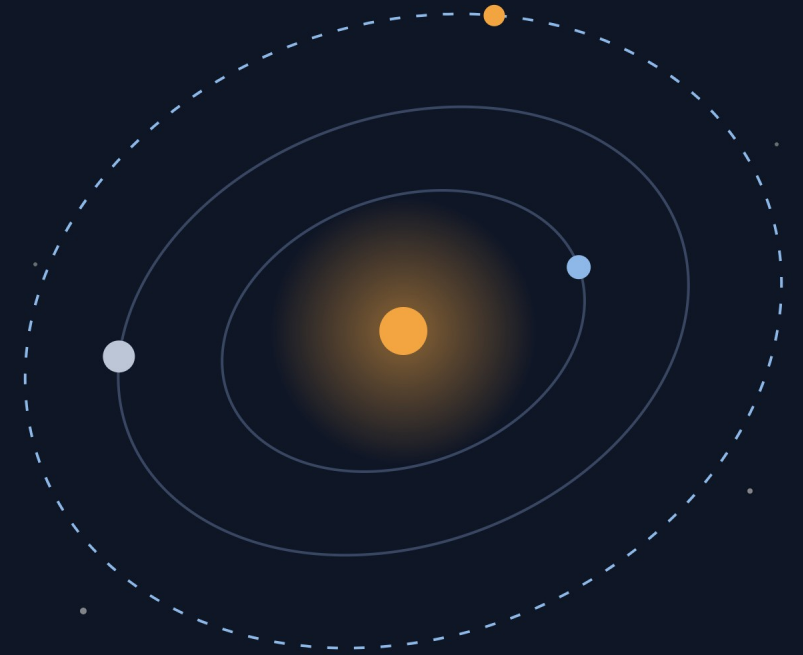


USING AI TO HELP WRITE RESEARCH CODE

Plan first, then compute

Why writing a short plan beats asking the AI for everything at once



01 · THE PROBLEM

Asking for everything at once

YOU TYPE

"Write a program that simulates a galaxy, runs it for 10 billion years and plots how fast the stars orbit."

YOU GET

Hundreds of lines of code that run and make a nice-looking plot...

...and no way to tell whether it is right.

Choices made for you

Which units, where the stars start, which shortcuts to take: the AI picks them without telling you.

Looks right is not the same as right

A small error can grow over billions of years and still give a believable plot.

Hard to find the mistake

If the answer is wrong, which part caused it? It is like a calculation where only the final answer is written down.

02 · THE FIX

What goes in the plan

A short plain-text file, for example PLAN.md, kept next to your code.

PLAN.md

Galaxy simulation

Goal

How fast do stars orbit at each distance?

Choices

Distance in light-years, speed in km/s

Stars start on circular orbits

Steps

[x] 1. One planet around one star

[] 2. Many stars around a centre

[] 3. Plot speed against distance

Notes

Oct 2: changed method, energy wasn't conserved

Goal

One clear question. If you cannot write it down, you are not ready to ask the AI.

Choices

Everything the AI would otherwise guess. You decide and write it down.

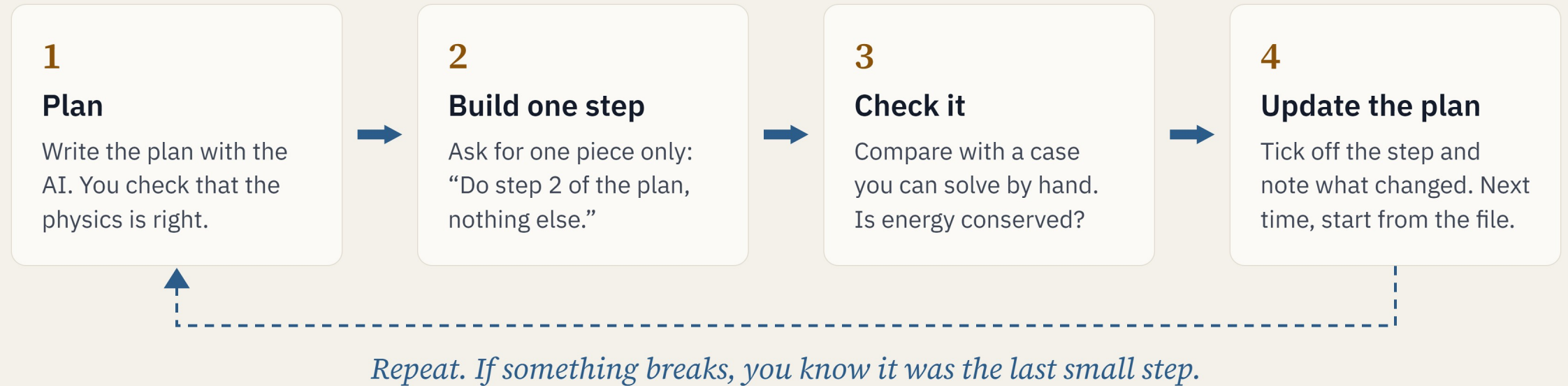
Steps

Small pieces, each one easy to check. Tick them off as you go.

Notes

Why things changed. The chat forgets; the file remembers.

Work in small steps



Write the physics before the code

Start with a plan file

Goal, choices, steps. Written before any code exists.

One step at a time

Small enough that you can read and understand every line.

Check every step

Test it against something you already know the answer to.

The file remembers

New chat? Start with “Read PLAN.md.” Later, it helps you write your thesis.

When asking for everything at once is fine: quick, throwaway jobs, like a single plot or converting units.